Recursion functions	Functions that calls itself is known as recursive.

Table – 7.1 – Python Functions and it's Description

7.2 Defining Functions

Functions must be defined, to create and use certain functionality. There are many built-in functions that comes with the language python (for instance, the print() function), but you can also define your own function. When defining functions there are multiple things that need to be noted;

- Function blocks begin with the keyword **"def"** followed by function name and parenthesis ().
- If any input parameters are present should be placed within these parentheses when you define a function.
- The code block always comes after a colon (:) and is indented.
- The statement **"return [expression]"** exits a function, optionally passing back an expression to the caller. A **"return"** with no arguments is the same as return None.

Note

Python keywords should not be used as function name.

7.2.1 Syntax for User defined function

```
def <function_name ([parameter1, parameter2…] )> :
        <Block of Statements>
        return <expression / None>
```

Note

In the above Syntax, the Text which is given in square bracket [] is optional.

Block:

A block is *one or more lines of code*, grouped together so that they are treated as one big sequence of statements while execution. In Python, statements in a block are written with *indentation*. Usually, a block begins when a line is indented (by four spaces) and all the statements of the block should be at same indent level.

Nested Block:

A block within a block is called nested block. When the first block statement is indented by a single tab space, the second block of statement is indented by double tab spaces.

Here is an example of defining a function;

```
def Do_Something( ):
    value =1          #Assignment Statement
    return value      #Return Statement
```


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